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The palaeomagnetism of major (Middle Proterozoic) igneous complexes, South Greenland and the Gardar apparent polar wander track

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ABSTRACT

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The Gardar Province comprises a major rift-related igneous suite emplaced in three pulses at ca. 1300–1280 Ma, ca. 1220 Ma and 1160–1125 Ma. This study reports a palaeomagnetic study of the northeastern sector of the present outcrop and incorporates igneous units belonging to the first and last episodes. Early lavas from the basal part of the rift infill record a reversal of magnetisation and define a palaeomagnetic pole at 207°E, 10°N (5 sites, dp/dm=7/14°). This primary magnetisation record has been partially overprinted in the aureoles of large alkaline complexes. The latter plutons include the North Qoroq intrusion (1291±61 Ma) yielding a pole position at 203°E, 13°N (65 samples from 12 sites, dp/dm=6/12°), the Motzfeldt intrusion (1291±31Ma) yielding a pole position at 212°E, 10°N (25 samples, dp/dm=6/12°), and the Narssarssuaq Stock yielding a pole position at 207°E, 9°N (23 samples, dp/dm=4/7°). The younger South Qoroq complex (1160±8 Ma) yields a pole position at 216°E, 42°N (67 samples from 9 sites, dp/dm=11/16°). Positive contact tests indicate primary cooling-related magnetisations. These bodies are predominantly of single polarity equivalent to W+ directions of magnetisations although reversals are recorded. The majority of a suite of NE–SW trending dykes emplaced at ca. 1154–1143 Ma are magnetised in a similar direction [D/I=283/51°, pole position at 231°E, 33°N (18 dykes, dp/dm=5/7°)] to the slightly-older South Qoroq complex.

Together with earlier results from the Gardar Province, these data define a motion of the region from shallow to equatorial and then back to intermediate palaeolatitudes between 1290 and 1140 Ma. Pole positions trace out an apparent polar wander path comprising a small loop followed by a long track. This track is the outward path of the "Great Logan" palaeomagnetic loop for which data from the Keweenawan Province of North America (ca. 1100–1050 Ma) define the return path. The loop was executed between ca. 1200 and 1050 Ma following a near-static interval between ca. 1290 and 1200 Ma. This long static interval was also characterised by constant field polarity. Asymmetry of geomagnetic field reversals is recognised in early Gardar events; it is not apparent in younger Gardar data although it recurs in Keweenawan times.

Introduction

Palaeomagnetic study of the long interval of Precambrian times (3800–590 Ma) is constrained by the uneven preservation of the geological record. The data come mostly from rather few major igneous events and thermotectonic episodes, plus rare volcanic-sedimentary infills of rifted zones (aulacogens). However, some of these events lasted for such protracted periods of time that the primary and secondary (overprinted) record can define long segments of apparent polar wander (APW, Piper, 1987). The well-studied Keweenawan Province of North America is an example. Since the early studies of DuBois (1962) it has been apparent that this province records a long APW track comprising a major segment of the "Great Logan" loop of Robertson and Fahrig (1971). Refinement of the palaeomagnetic re-

cord from this province (Halls and Pesonen, 1983) shows that only the return path of the loop (now referred to as the "Keweenawan Track") is recorded here. The extrapolation of this APW path backwards in time to ca. 1200 Ma, a period well defined by results from basic igneous suites, has remained unclear (see review by Roy, 1983).

The Gardar Igneous Province of South Greenland, which forms the subject of this paper, is an example of a major rift-related igneous episode of Middle–Upper Proterozoic age. Whilst igneous activity was probably concentrated in three short pulses (Blaxland et al., 1978), the collective palaeomagnetic results identify in outline a consistent pattern of APW between ca. 1300 and 1150 Ma (Piper, 1977a). When combined with data spanning the same interval from adjoining shield areas, they identify an APW track which appears to be the precursor of the path recognised in the Keweenawan Province. Collectively these data define the pattern of continental movement over some 150 Ma of Proterozoic times (Piper, 1987).

Previous investigations of the Gardar Province have comprised studies of an early lava succession (Piper, 1977a) which is currently being extended (Thomas and Piper, 1991; Thomas, in prep.) and late giant dykes and central intrusive complexes (Piper, 1977b, c). Results from dyke swarms provide a measure of APW between the middle and late stages of the igneous activity (Piper and Stearn, 1977). This paper reports an investigation of major intrusive complexes emplaced during the early and late stages of Gardar magmatism, and of dykes emplaced close to the younger event. Complementary results are also reported from the base of the early-formed volcanic infill to the Gardar Rift.

Geological framework

A regional description of the Gardar Province is given by Emeleus and Upton (1976) and is updated, including petrologic data, by Upton and Emeleus (1987). The sequence of major events is summarised in Table 1. Absolute age determinations are based mainly on the Rb–Sr method and summarised by Blaxland et al. (1978). These ages, adjusted to revised decay constants (Steiger and Jäger, 1977), are quoted here.

The province is developed within a deeply-eroded granitic basement of Middle Proterozoic age (van Breemen et al., 1974). The earliest stage is recorded by up to 3000 m of clastic sediments and lavas composing the Eriksfjord Formation (Poulsen, 1964, the la-

TABLE 1

GARDAR IGNEOUS PROVINCE, SOUTH GREENLAND

Western Sector :	Eastern Sector :
(LATE PHASE)	
Sparse ENE dykes (Dolerite, Lamprophyre, Trachyte and Phonolite)	
Tugtutôq Central Complex (1124)	Klokken Complex (1142 ± 11)
Nunarssuit Syenite (1137 ± 14)	
Nunarssuit Granite (1144 ± 21)	
Helene Granite (1132 ± 31)	
Alangorssuaq Gabbro (1126 ± 48)	*Ilímaussuaq Complex (1143 ± 21)
	Igdlerfigssalik Complex (1154 ± 15)
**ENE Dyke Swarms (Dolerite, Trachyte, Phonolite, Rhyolite, "Big Feldspar" Dykes)	
*Tugtutôq Giant Dykes, Narssaq Complex	
*Hviddal Giant Dyke (1154)	
	**South Qoroq Complex (1160 ± 8)
(MIDDLE PHASE)	
*Kungnat Complex (1219 ± 17)	
Ivigtut Complex (1222 ± 25)	
*Dolerite Dykes (Mainly WSW-ENE to WNW-ESE) ca.1280-1219	
*Lamprophyre Dykes	
(EARLY PHASE)	
	**North Qoroq Complex (1291 ± 61)
Gronnedal-Ika Complex (1299 ± 17)	**Motzfeldt Complex (1291 ± 31)
GARDAR RIFT :	*4. Upper Lavas (Aphyric and feldspar-phyric basalts and upper trachytes)
	3. Upper Sediments (mainly drab to red quartzites)
	*2. Middle Lavas (Basalts)
	*,**1. Lower Sediments (mainly red sandstones and siltstones) and Basaltic Lavas
Ketilidian Basement (incl. the Julianehab Granite) 1850-1600	

(** = This Study; * = Previous palaeomagnetic studies)

vas and sediments of the Early Phase in Table 1). Lavas are olivine basalts erupted rapidly in a subaerial environment with little or no erosion before extrusion of succeeding units. At Qagssiarssuk however, at the base of the succession (palaeomagnetic sites 48–57, Fig. 2) local emplacement of ultrabasic and carbonatite intrusions was associated with explosive volcanism. Lavas from the bottom of the Eriksfjord Formation overlying granitic base-

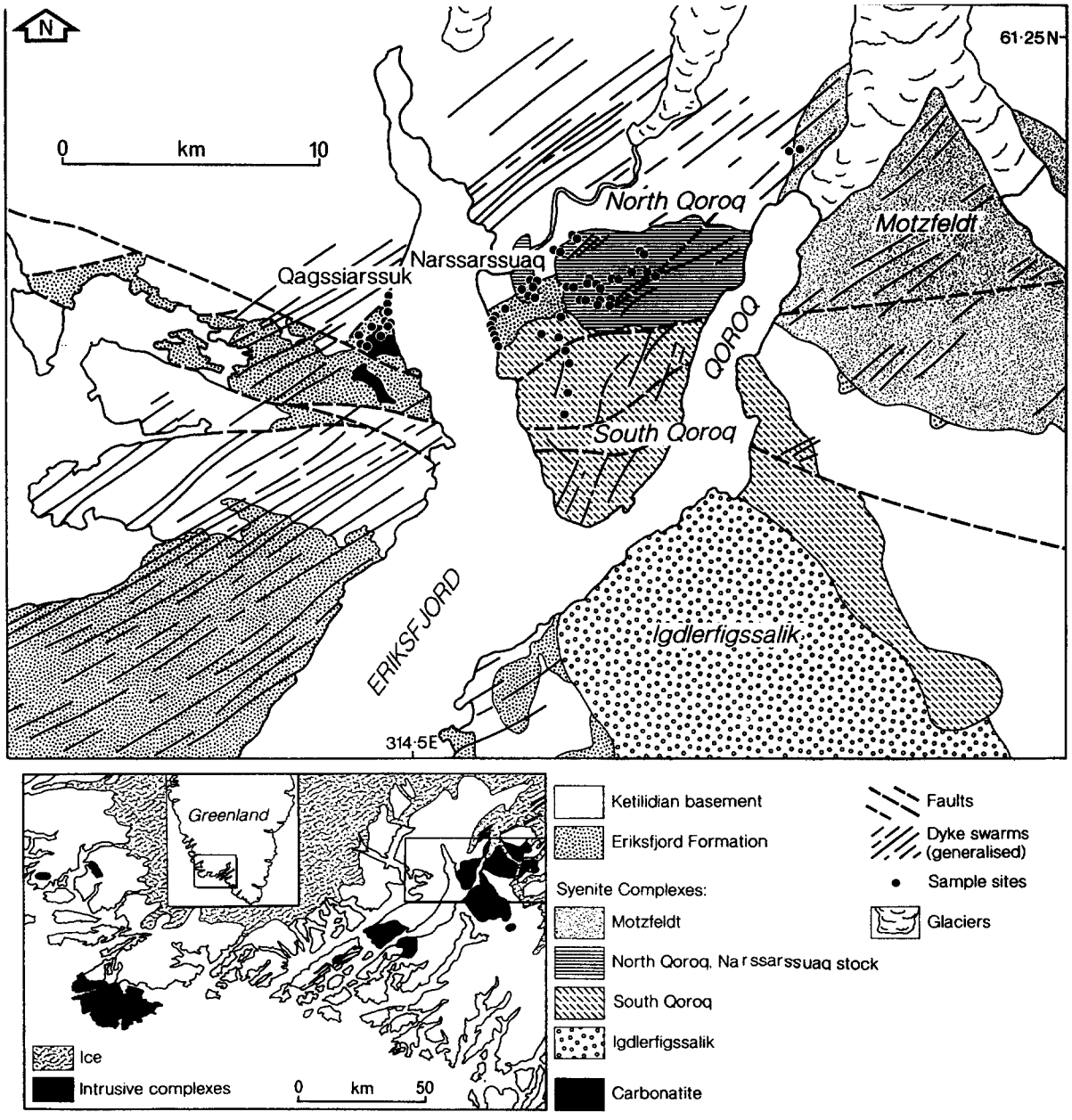


Fig. 1. Generalised geology of the northeastern sector of the Gardar Igneous Province, South Greenland, composing the subject of this study. The small-scale map shows the regional location of this area and the distribution of the major intrusive complexes within the province.

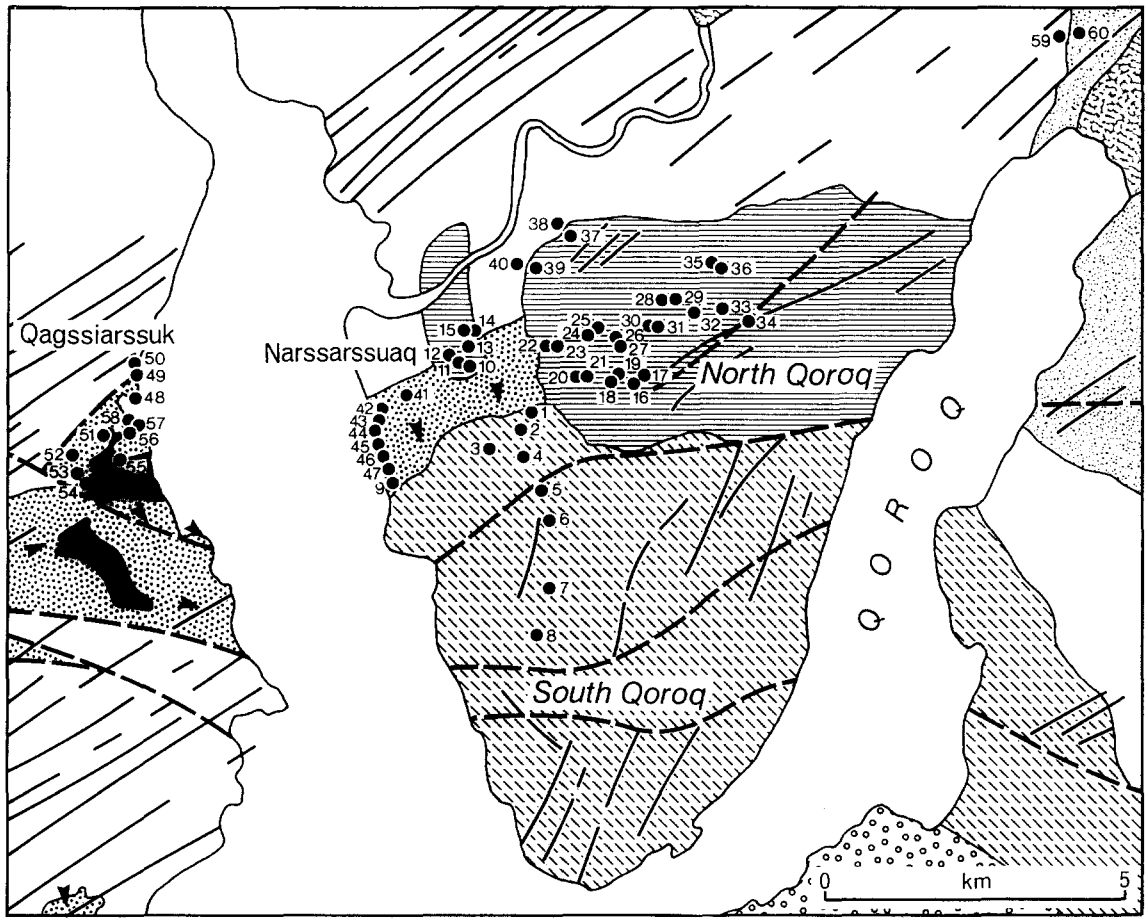


Fig. 2. Palaeomagnetic sampling sites of this study. The rock symbols are the same as for Fig. 1. The arrows are directions of geological tilt.

ment have also been sampled at Narssarssuaq (sites 42, 45–47) in the vicinity of large younger syenite bodies (Fig. 2). Outliers of the Eriksfjord Formation are cut by the Motzfeldt Complex (1291 Ma) and the age of deposition is currently constrained between this time and the age of final cooling of the underlying basement (~ 1600 Ma, van Breemen et al., 1974).

In the north eastern sector of the Province not covered by the inland ice (Fig. 1) a cluster of intrusions comprises four large alkaline complexes (Motzfeldt, North Qoroq, South Qoroq and Igdlerfigssalik, Table 1 and referred to as the "Igaliko Group") plus a number of satellite bodies including the Narssarssuaq stock of this study. They probably decrease in age from the NE to the SW although the relationship between the Motzfeldt and North Qoroq centres is not established. The Motzfeldt complex (1291 ± 31 Ma) is an annular complex composed of several nepheline syenite members (Emeleus and Harry, 1970), Only the margin of the complex exposed on the western margin of Qoroq Fjord (Fig. 1) is accessible by foot where sites 60 and 59 are in syenite and a dolerite dyke respectively at the contact (Fig. 2). The North Qoroq centre (1291 ± 62 Ma) is a nepheline syenite

complex composed of 5 concentric intrusions truncated by later syenites of the South Qoroq centre (Fig. 1). The latter body (1160 ± 8 Ma) is also composed of five concentric nepheline syenite divisions. On the SE side of Eriksfjord this body is cut by the Igdlerfigssalik complex (1154 ± 54 Ma) which was not sampled in this study. Sampling sites in these bodies are numbered in Fig. 2.

The earliest ("BDO") dyke swarms have E–W to WNW–ESE trends but are scarce in the NE sector comprising this study area. The most prominent dykes here have a NE–SW to ENE–WSW trend and are assigned to "BD1" and "BD2" groups (within the Middle and Late Phases of Table 1). This is a varied assemblage comprising several giant dykes, ultrabasic dykes, dykes with large feldspar phenocrysts and/or anorthosite inclusions and a range of more silicic types including trachytes. Twenty one dykes have been sampled (Fig. 2) where they cut the other units studied. Members of this suite cut the Older (Hviddal) Giant Dyke dated 1154 ± 9 Ma but are cut by the Ilímaussaq intrusion (the intrusion outcrop SW of the Erikfjord Formation in Fig. 1) dated 1143 ± 21 Ma. They also cut the Motzfeldt and North Qoroq bodies, whilst the South Qoroq body is only apparently cut by the youngest NNE–SSW trending (BD2) members with sinuous orientations suggestive of emplacement into a semi-ductile medium. Hence it seems likely that most of these dykes were intruded over a short interval of time at ~1150 Ma.

Tectonic deformation is restricted to faulting of syn- or post-Gardar age. Possibly as a result of block faulting of late Gardar age, the Eriksfjord Formation lavas dip at angles of 15° or less to the S and SW (Fig. 1). SSE–NNW trending faults cut rocks older than the youngest part of the Igdlerfigssalik Complex and long E–W trending faults cut the entire region (Emeleus and Upton, 1976). Apart from minor volcanism associated with rifting in Mesozoic times, and uplift resulting in seaward tilting of the coastal margin in Tertiary times, the region has been geologically inactive since Gardar times.

Palaeomagnetic studies

Cores were sampled with a petrol-driven motorised drill and oriented by sights on the Sun and magnetic bearings in conjunction with topographic sights. An abundance of fresh glaciated exposure permitted an equitable spacing of cores (typically numbering 7–10) over several metres of outcrop at each site. In the laboratory samples were cut into 2.4 cm long cylinders. Magnetisations were measured before and after alternating field (a.f.) and thermal treatment by 'minispin' and parastatic magnetometers. Cleaning steps by the a.f. method were 5 milliTesla (mT) to 30mT and then 10mT to peak fields of 140 mT. Cleaning steps by the thermal method were 100°C to 500°C and then in steps of 20 or 25°C to the Curie point of the magnetite (580°C) or hematite (680°C) remanence carriers. Following examination of orthogonal vector projections, components of magnetisation were resolved by principal component analysis (Kirschvink, 1980; Sherwood, 1989). With the exception of the lavas where the thermal method provided better component resolution, both methods of treatment produced component definitions of comparable quality.

Both earlier and later generations of the syenites have similar intensities of magnetisation (Fig. 3). In addition to trachytes, the NE dyke suite includes basaltic members and some magnetisations are correspondingly stronger. The lavas, although of basaltic composition, have lower intensities because the lava pile has undergone a low grade hydrothermal metamorphism not experienced by the later dykes (Piper, 1977a). Within the aureole zones of the major plutons the remanence in the lavas has been reconstituted (next subsection) and the intensities of magnetisation tend to be higher (Fig. 3).

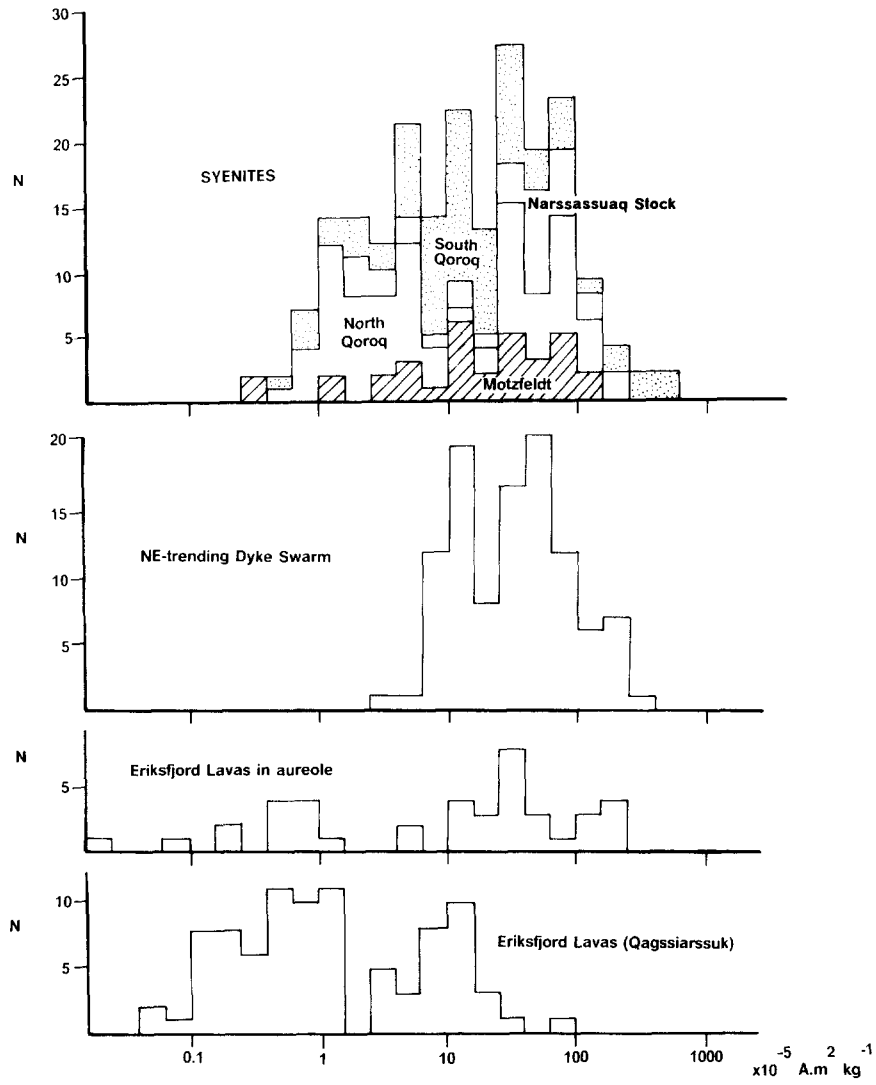


Fig. 3. Normalised distribution of intensities of magnetisation ($\times 10^{-5}$ A m^2 kg^{-1}) of the igneous rock suites of this study.

Basal lavas of the Gardar Rift (Eriksfjord Formation)

The palaeomagnetic sites in these rocks remote from later igneous activity (with the exception of minor dyke emplacement) come from Qagssiarssuk on the western side of Eriksfjord (Fig. 2, sites 48, 50, 52, 54-57). They comprise an outlier of the Gardar lava-redbed succession exposed to the SW (Fig. 1). The basal lavas are dominated by single high blocking temperature components only fully subtracted above the Curie point of magnetite (Sample 48-6 in Fig. 4) and therefore resident, at least in part, in hematite. The westerly shallow positive directions are compatible with directions resolved from the bulk of the lavas within the Lower Sediments of the main succession (Table 1) which comprise a major outcrop to the SW (Fig. 1, Piper, 1977a; Thomas and Piper, 1991). However, the two highest members of the thin south-easterly dipping succession at Qagssiarssuk (sites 56 and 57) have ENE directed magnetisations of

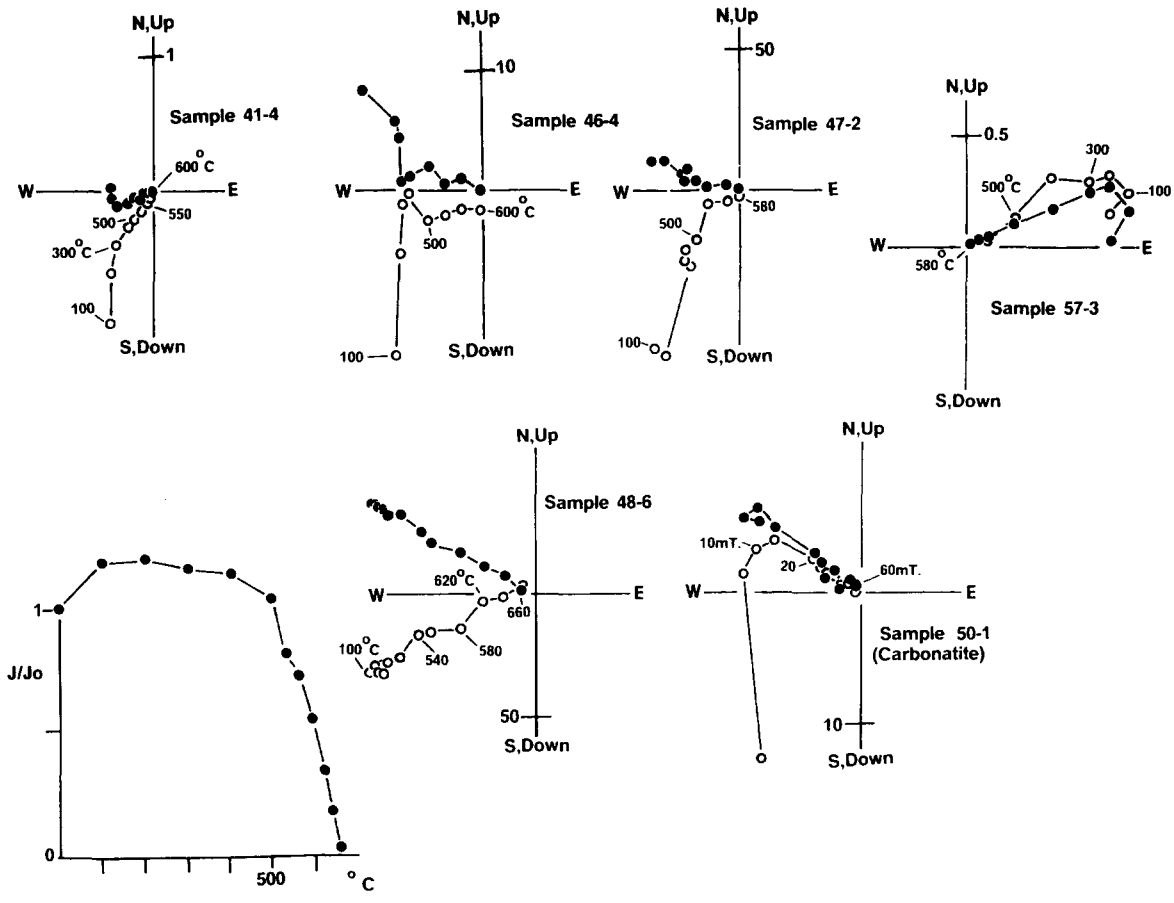


Fig. 4. Representative results of progressive thermal and alternating field demagnetisation of basal lavas from the Eriksfjord Formation at Qagssiarssuk and within the aureole of the South Qoroq intrusive complex at Narssarssuaq. The palaeomagnetic vector is projected onto orthogonal horizontal (solid symbols) and vertical (open symbols) planes to illustrate component behaviour and the figures on the axes are $\times 10^{-5}$ A m^2 kg^{-1}. Directions are shown in situ.

shallow inclination residing in a lower blocking temperature magnetite phase (eg. sample 57-3 in Fig. 4). This does not represent a precise reversal of the palaeofield because the antiparallel declinations differ by ca. 4° and the inclinations by ca. 10° (Table 2). It may reflect, in part, the asymmetry of field reversal recorded at higher levels in the Gardar succession (Thomas and Piper, 1991) but reversal at these localities is incomplete. Two isolated shore exposures 2 km SE of Qagssiarssuk, which include pillow lava and are evidently higher in the succession than sites 56 and 57, record asymmetric reversals [site 1 (D/I=89/39°) and site 2 (D/I=77/20°) of Piper, 1977a]. A reversed unit has since been located at the base of the Lower Group lavas in the main succession to the SW (D.N. Thomas, pers. commun.). Hence a double reversal of the palaeofield (W→E→W) is recorded in the basal part of the Gardar succession. The contrasting demagnetisation characteristics between the two polarities (Fig. 4) are commensurate with the lower magnetite oxidation states observed in the E-directed polarity lavas by Piper (1977a).

The carbonatite (55) and tuffsite pipe (54) at Qagssiarssuk have magnetite-resident magnetisations (eg. sample 50-1 in Fig. 4) of similar declination, but shallower inclination than

TABLE 2

Palaeomagnetic results from basal lavas of the Eriksfjord Formation, Gardar Igneous Province, South Greenland

Site No.	N/n	R	k	α_{95}	D°	I°	D° (tilt adjusted)	I°
(i) Narssassuaq district:								
42	10/8	7.54	15	15	277	50*	259	56
44	6/2	1.99	220	17	269	38*	257	42
45	6/5	4.98	176	6	271	39*	259	45
46	8/5	4.72	14	21	284	32*	274	40
47	8/8	7.62	18	13	281	47*	265	54
(ii) Qagssiarssuk district:								
48 (i)	7/7	6.83	35	20	288	5	287	10+
(ii)	7/7	6.80	30	11	292	45*	287	50
50	7/7	6.55	13	17	303	−15	304	−9+
52	8/8	7.93	97	6	283	1	283	5+
54	8/3	2.76	8	46	294	8	293	13+
55	12/11	10.77	42	7	281	−18	282	−13+
56	6/3	2.97	80	14	68	−22	65	−22
57	6/5	4.72	14	21	55	−24	52	−23
Group Mean Results:								
Lavas, tilt adjusted result (+):								
	5 sites	4.87	31	14.1	292	3	290	1
							Palaeomagnetic Pole	
							E	N dp/dm
							207.4	9.8 7.0/14.1
Lavas, late Gardar overprints (*):								
	6 sites	5.94	78	7.6	279	42	267	48
							Palaeomagnetic Pole	
							E	N dp/dm
							229.3	25.3 5.8/9.4

Sites 50 and 55 are in carbonatites; site 54 is in a tuffsite pipe. Directions employed in calculations of inferred primary and overprinted components are indicated (+) and (*) respectively. R is the resultant vector derived from n samples taken from a site population of N; k is the corresponding Fisherian precision parameter [$=(n\text{-}1)/(n\text{-}R)$] and α_{95} is the radius of the cone of 95 percent confidence about the mean direction. D and I are the declination and inclination of the mean remanence. dp and dm are the semi axes of the oval of 95% confidence about the pole position in the colatitude direction and at right angles to it respectively.

the (W+) lavas. This may either record a minor difference in emplacement age or reflect a post-eruptive origin for the hematite-held remanence in the lavas.

On the east site of Eriksfjord the basal lava succession lies within 1.5 km of the contact of the South Qoroq pluton and degrees of magnetic overprinting are always present. In the majority of samples remagnetisation is complete and a component with westerly positive direction similar to the South Qoroq body (last subsection of this section) is subtracted at the Curie point of magnetite (sample 41-4, Fig. 4). In other samples two component behaviour is recognised with this W+ component removed by about 500°C to recover a shallow westerly-directed component between 500 and 600°C similar to the predominant direction recognised elsewhere in the lower Gardar lavas (samples 46-4 and 47-2 in Fig. 4). The tilt-adjusted mean of these shallower directions (D/I=287/26°, $n=6$, $R=5.5$) is of similar decli-

TABLE 3

Site and group mean palaeomagnetic results, North Qoroq in intrusion and Nassarssuaq Stock, Gardar Igneous Province, South Greenland

Site No.	N/n	R	k	α_{95}	D°	I°
(i) Nassarssuaq Stock						
11	8/7	6.89	53	8	283	−4
12	9/9	8.55	18	13	286	−1
13	7/7	6.63	16	15	296	7
Group Means Results:						
3 sites		2.98	84	14	288.2	0.8
			Palaeomagnetic Pole			
			°E	°N	dp/dm	
			209.0	9.0	6.8/13.6	
23 samples		21.90	20	7	288.0	0.6
			Palaeomagnetic Pole			
			°E	°N	dp/dm	
			207.2	8.8	3.5/ 6.9	
NW-component sites 12 & 13						
4 samples		3.86	22	20	307	−34
(Palaeomagnetic Pole 184°E, O°S, dp°/dm=13/23)						
(ii) North Qoroq Intrusion						
16	7/6	5.75	20	16	293	4
17	7/5	4.55	9	27	308	4
22	6/3	2.91	23	26	90	−15
23	6/5	4.73	15	21	285	4
25	7/7+	6.93	87	7	285	37
27	7/4	3.79	14	25	319	−4
29	6/5	4.88	34	13	268	10
31	9/8	7.68	22	2	313	−3
32	5/5	4.94	74	9	313	−1
34	7/6	5.84	30	12	312	5
36	7/6	5.71	17	17	300	24
37 (a)	7/6+	5.96	111	6	44	57
(b)	7/5	4.63	11	24	296	−3
40 (contact)	7/7	6.90	58	8	269	−18
Group Mean Results:						
12 sites		11.25	15	12	295.7	2.5
			Palaeomagnetic Pole			
			°E	°N	dp/dm	
			202.6	13.2	5.9/11.7	
65 samples		57.50	9	6	297.4	1.6
			Palaeomagnetic Pole			
			°E	°N	dp/dm	
			200.8	13.5	3.2/6.4	
(iii) Motzfeldt Intrusion						
25 samples		21.43	6	12	286.0	4.3
			Palaeomagnetic Pole			
			°E	°N	dp/dm	
			211.8	9.6	6.1/12.2	
(Late Gardar Overprint)						
8 samples		7.81	37	9	311.1	50.9
(Palaeomagnetic Pole 204.6 E, 46.8 N dp/dm=8.4/12.5)						

Footnote: Symbols are as for Table 2. + Sites excluded from group mean calculations.

nation and somewhat steeper inclination than the lava remanence at Qagssiarssuk suggesting a residual contamination by the overprint; it is also possible that this component has been modified because it has not positively been identified into the hematite range like the W+ components at the latter locality (Table 2). The polarity implies that the lavas immediately overlying the basement at Narssarssuaq are part of the same succession as the lower members on the west side of the fjord.

The North Qoroq Complex

This body has a characteristic remanence with shallow W to NW directions (Fig. 5). It is magnetised predominantly with one polarity but a reversal of the palaeofield is recognised at one site (22 in Table 3) and in isolated cores from other sites (e.g. sample 29-5 in Fig. 5). Remanence acquisition during protracted cooling is therefore indicated. An amphibolite lens in granitic basement close to the contact with the pluton has a high and narrow blocking temperature component residing in magnetite (sample 40-1, Fig. 5); the direction of this component is close to the characteristic remanence in the pluton (site 40, Table 3) supporting the view that the later records a primary cooling event.

Motzfeldt Complex margin

Samples from the Motzfeldt body come from

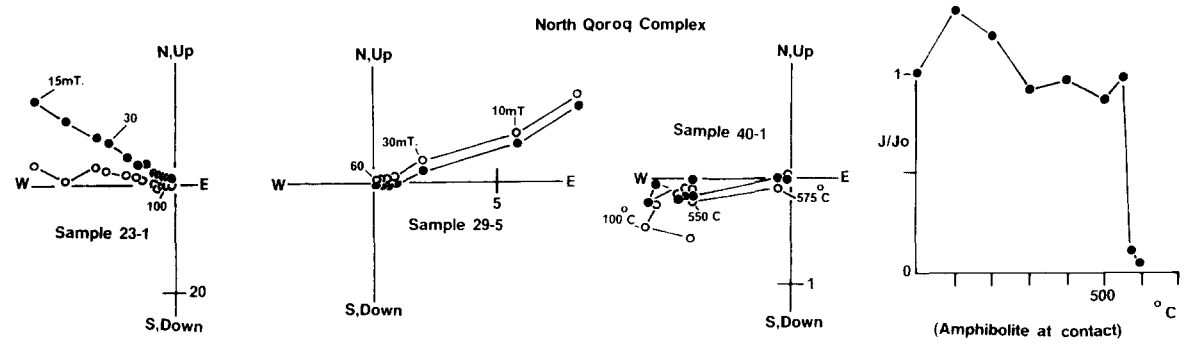


Fig. 5. Demagnetisation results from samples of the North Qoroq intrusive complex and an amphibolite lens close to the contact with the pluton. Symbols are as for Fig. 4.

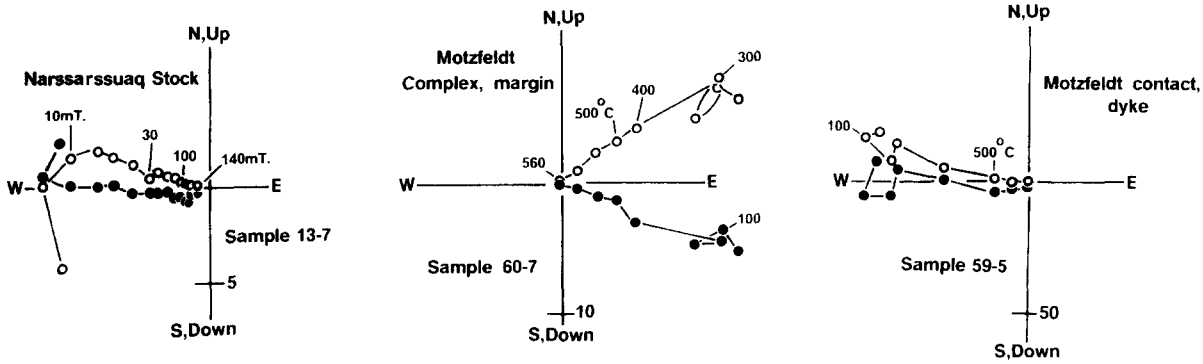


Fig. 6. Demagnetisation results from samples of the Narssassuaq Stock and the Motzfeldt contact and margin. Symbols are as for Fig. 4.

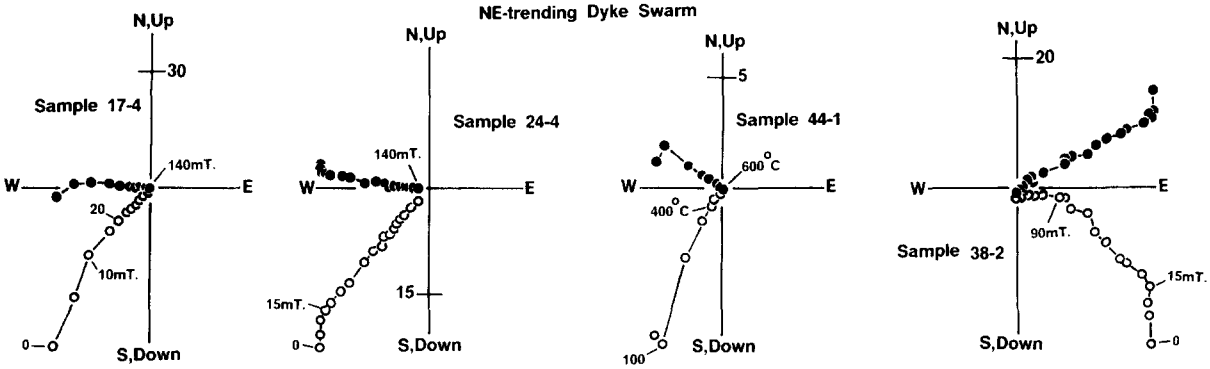


Fig. 7. Palaeomagnetic results from samples of members of the NE-trending dyke swarm. Symbols are as for Fig. 4.

the western margin and cannot be considered representative of the whole pluton. The remanence is dominated by single components residing in magnetite with easterly or westerly declinations and shallow inclinations (Fig. 6). A reversal of the palaeofield has evidently been recorded here by prolonged cooling (samples 59-5 and 60-7 in Fig. 6 and Fig. 9a). A remanence with similar direction is present in an E–W trending BDO dyke cut by the pluton at its contact (sample 59-5 in Fig. 6). The mean direction derived from 25 samples (D/I=286/

4°, Table 3) is not significantly different from the mean direction of remanence in the North Qoroq complex supporting a similar age of magnetisation and commensurate with the Rb–Sr isochrons from these intrusions (Blaxland et al., 1978 and Table 6).

Narssarssuaq Stock

Although this pluton is undated it has similar remanence characteristics to the nearby North Qoroq Complex (Fig. 2). Following initial removal of a viscous component, single convergent vectors with shallow westerly directions are isolated and evidently resident in magnetite (sample 13-7 in Fig. 6). All samples have the same polarity. The mean direction (D/I=288/1°, Table 3) is close to the characteristic remanence in the Motzfeldt and North Qoroq complexes. A similar age of remanence is therefore implied.

NE–SW trending Dyke Swarm

The great petrological diversity of these dykes is not reflected in a varied magnetic imprint. With few exceptions, they are characterised by remanences of high stability with W to WNW intermediate positive directions (samples 17-4, 24-4 and 44-1 in Fig. 7) contrasting with the remanence in all of the country rocks except in the younger South Qoroq Complex (next subsection). Only one polarity is represented (Table 4). It is therefore likely that they were all emplaced and magnetised over a short interval of time. This interval is constrained to ca. 1154–1145 Ma by Rb–Sr ages on the bodies cutting, and cut by, these bodies (Blaxland et al., 1978 and Table 6). At two sites (38 and 39) multi-component behaviour is present (eg. 38-2 in Fig. 7) in divergent E to NE positive directions. This is not the record of a complete reversal of the palaeofield but the directions are comparable to a minority of directions recognised in dykes (6 out of 62) from the same swarm 50–100 km to the south west (Piper and Stearn, 1977). The mean of these shallow inclination sites (D/I=82/4°, N=8, R=7.72, pole position at 51°E, 6°N, dp/dm=6/11°) is similar to results from the reversed component of remanence in the Eriksfjord lavas. Whilst the NE–SW trending swarm is linked to emplacement of the Tugtutòq Central Complex in the late phase of Gardar activity (Upton et al., 1990), this result suggests that representatives of an early episode of dyke

TABLE 4

Site and Group Mean Palaeomagnetic Results, NE trending basic dyke swarms, Gardar Igneous Province, South Greenland

Site No.	*N/n*	*R*	*k*	α_{95}	D°	I°
(i) Nassarssuaq district						
10	10/10	9.79	43	8	280	48
17*	5/ 5	4.99	304	4	276	43
18 (contact)	3/3	2.94	34	21	288	58
19*	7/6	5.99	435	3	302	69
20	6/6	5.87	38	11	280	53
21	5/5	4.98	196	6	282	38
24	6/6	5.99	379	4	280	49
26	4/4	3.99	603	4	283	61
28	7/7	6.91	70	7	268	54
30	7/7	6.98	290	4	277	51
33	7/7	6.97	199	4	276	60
35	6/6	5.98	331	4	260	64
38	6/5+	4.89	36	13	67	4
39 (i)	6/3+	2.92	25	25	29	24
(ii)	6/2+	1.93	19	–	90	25
41	6/6	5.96	122	6	274	42
43	6/3	2.97	68	15	282	44
44	6/5	4.89	36	13	298	51
(ii) Qagssiarssuk district						
49	6/4	3.94	48	14	309	48
51	6/5	4.96	111	7	279	43
53	7/7	6.79	28	12	295	31
Group Mean Result: 18 sites		17.63	46	5	282.9	50.9

Palaeomagnetic Pole		
°E	°N	dp/dm
230.8	33.4	4.7/6.9

+ Directions excluded from site mean calculation.
*Trachyte dykes. Symbols are as for Table 2.

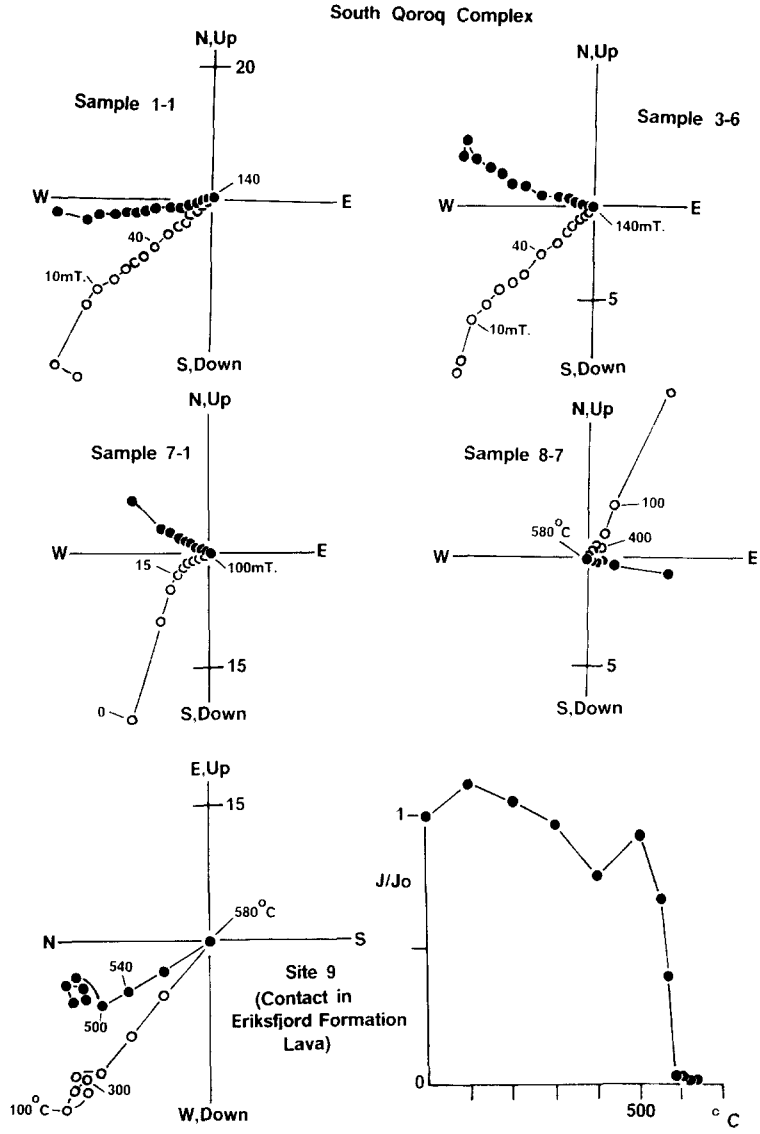


Fig. 8. Palaeomagnetic results from samples of the South Qoroq complex and contact. Symbols are as for Fig. 4.

injection paralleling the Gardar rift (Fig. 1) may be sparsely represented here. Alternatively they may record a sporadic overprinting event at a much later time (cf. next section and Fig. 11).

The South Qoroq Complex

The syenites of the South Qoroq complex exhibit high stabilities to progressive demagnetisation. Following subtraction of a steep component at initial steps, single convergent vectors were isolated by both cleaning methods (Fig. 8). These vectors have intermediate positive inclinations throughout the traverse into the intrusion. They are predominantly of one polarity although a reversal is detected (sample 8-7 in Fig. 8). The declinations tend to swing around from W to NW into the interior of the pluton (Table 5). This is illustrated in Fig. 8 by comparison of typical samples from sites 1, 3 and 7. The sites with declinations

TABLE 5

Site and Group Mean Palaeomagnetic Results for the South Qoroq Intrusion, Gardar Igneous Province, South Greenland

Site	*N/n*	*R*	*k*	α_{95}	D°	I°
1	7/7	6.89	52	8	275	47
2	7/7	6.86	43	9	275	58
3	8/8	7.70	23	12	298	47
4	7/6	5.80	26	14	265	54
5	9/8	7.80	35	10	346	43
6	8/8	7.72	25	11	304	66
7	7/7	6.86	44	9	303	41
8	7/5	4.77	17	19	315	45
9+	14/11	10.72	36	8	308	40
Group Mean Results:						
9 sites		8.60	20	12	299.9	51.2
			Palaeomagnetic Pole			
			°E	°N	dp/dm	
			215.9	41.8	10.8/15.9	
67 samples		62.30	14	5	301.2	50.7
			Palaeomagnetic Pole			
			°E	°N	dp/dm	
			214.3	42.1	4.4/6.5	

Footnote: D and I are the mean declination and inclination derived from n samples of a site population N. R is the magnitude of the resultant vector, k is Fisher's precision parameter [$=(n-1)/(n-R)$] and α_{95} is the radius of the cone of 95 percent confidence about the mean direction. dp and dm are the radii of the oval of confidence about the pole position in the colatitude direction and at right angles to it respectively. + Contact site in Eriksfjord Lavas.

> 300°E come from the interior unit ("SQ3") of Emeleus and Harry (1963) while sites with declinations < 300°E are in the marginal unit ("SQ2") of these authors.

As a result of this effect, the distribution of site mean declinations is twice as large (50°) as the distribution of inclinations. This effect could result from (i) block rotations about fault planes cutting the sampled traverse (Macdonald, 1980 and Fig. 2), or (ii) migration of the palaeofield direction during the interval of cooling of the pluton.

A single site (9) in Gardar lavas at the contact with the western margin of the body shows no vestige of the typical shallow W+ direction found in the lavas in the aureole but more remote from the contact (first subsection of this section). A single magnetite-resident component at this site has a NW positive direction closest to the sites 3, 7 and 8 (Fig. 8, Table 5). This does not resolve between possibilities (i) and (ii) above and detailed interpretation is complicated by possible faulting along the contact at this point. If (i) is an important consideration, it would be expected to contribute to comparable, or greater, dispersions of directions in the older plutons. The North Qoroq complex however, illustrates similar dispersions of declination and inclination (Fig. 9b) with a somewhat smaller dispersion of site declinations than the South Qoroq complex (Table 3). Furthermore the largest change in declination is observed between sites 4 and 5 across a large fault crossing the pluton (Fig. 2) and could be accommodated by a clockwise rotation of the block in the south (hanging wall) side. With the exception of these two sites, the migration of the palaeofield direction becomes progressively more northerly into the intrusion within confidence limits (Table 5). The same clockwise migration of remanence declination is also recognised in the overprint in the aureole sites (cf. sites 42 to 47 in Table 2 and site 9) as the contact is approached. Together these observations favour (ii) as the dominant mechanism for this dispersion. A comparable field migration also appears to be recorded during the cooling interval of the slightly-younger Ilímaussaq body (~1140 Ma, Table 6) 35 km to the SW (Piper, 1977b and Fig. 1, inset).

The group mean direction for the South Qoroq body (D/I = 300/51°) is also rotated slightly clockwise from the mean remanence (D/I = 283/51°) in the NE–SW trending dykes. Although the complex is cut by some of these dykes, it is likely to have consolidated and acquired its magnetisation slightly later than most of the dykes (second section).

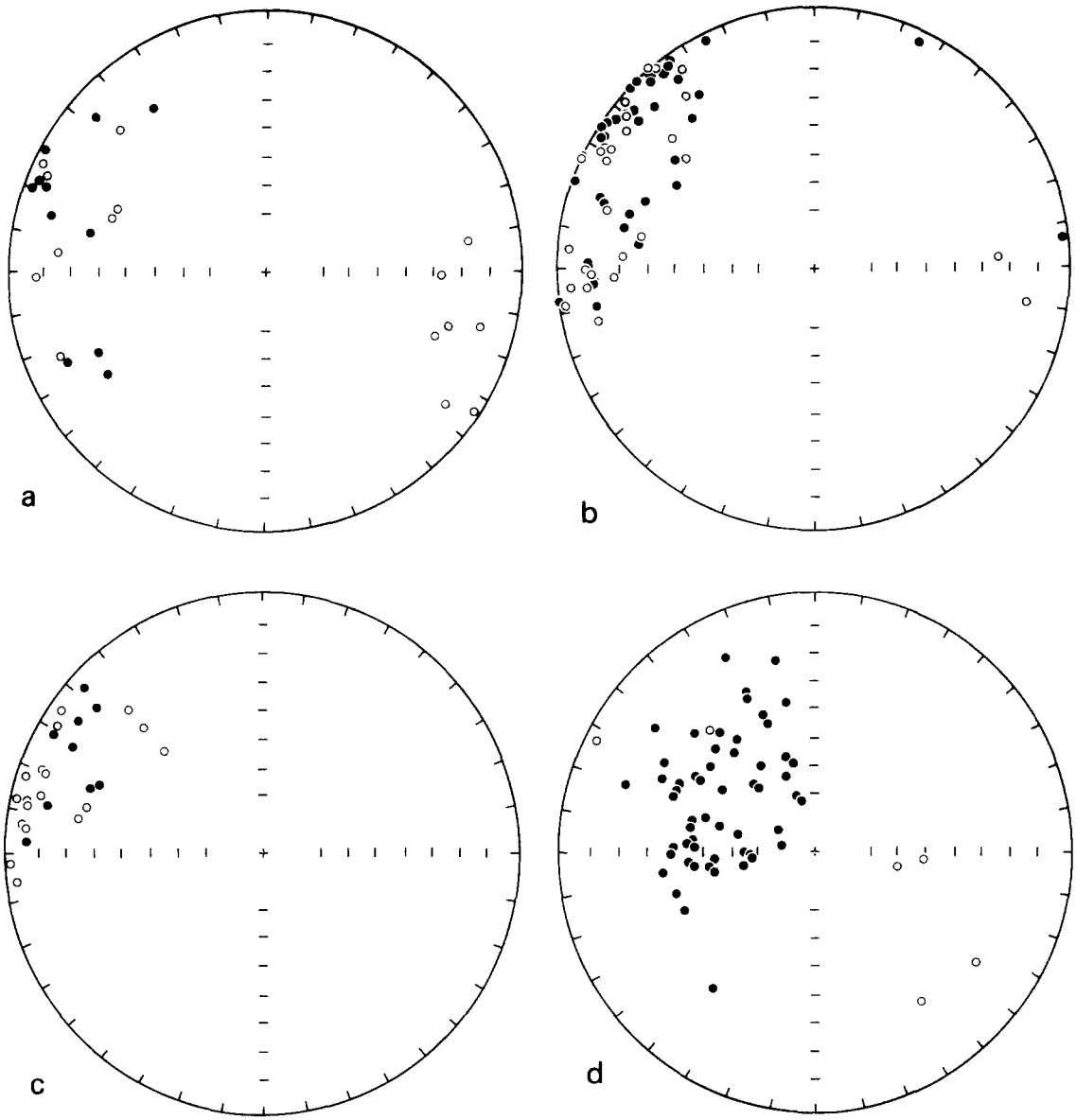


Fig. 9. Characteristic magnetisations (in situ) from samples in (a) the Motzfeldt margin and contact, (b) the North Qoroq Complex, (c) the Narssarssuaq Stock and (d) the South Qoroq Complex. Equal area projection.

Discussion

These new palaeomagnetic results from the Gardar Province provide a number of pole positions for the Laurentian Shield from the interval ~1300–1150 Ma, with the characteristic magnetisations in four units (Motzfeldt, North Qoroq, South Qoroq and the majority population of NE dykes) being temporally-constrained by Rb–Sr isochron ages. The remaining units have ages delimited by geological relationships. The collective data are summarised in Table 6.

Identification of an asymmetric reversal at the base of the Gardar succession extends the record of this phenomenon throughout this sequence. Eight successive asymmetric reversals

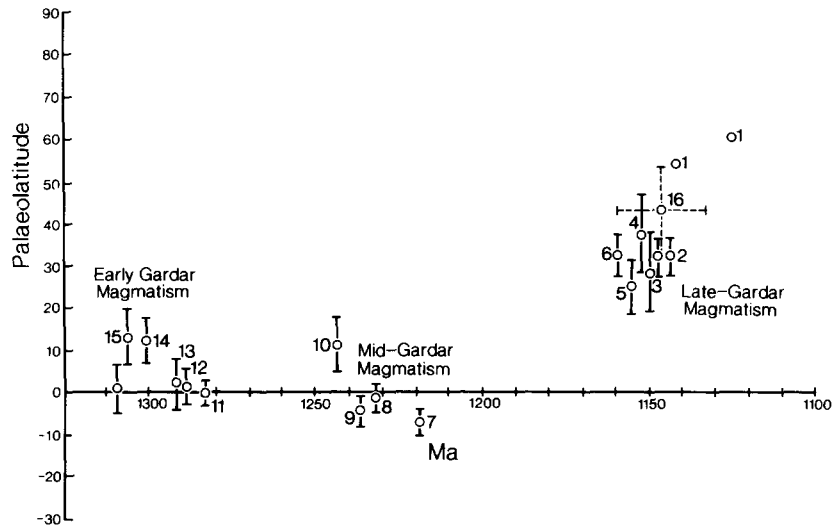


Fig. 10. Palaeolatitudes of South Greenland during Gardar times deduced from the palaeomagnetic results from this province. The data are summarised in Table 6 and numbers refer to the results listed. For this assessment the Eriksfjord Formation is assumed to be slightly older than the earliest intrusive complexes. Result 16 is from redbeds at the base of the Eriksfjord succession of the same geological age as the early Gardar magmatism. This is therefore interpreted as a remanence fixed by diagenesis during the late Gardar magmatic phase (Table 1).

are now recognised during the total lava succession recorded in the Eriksfjord Formation (first subsection of previous section, Piper, 1977a; Thomas and Piper, 1991); the anomalous directions in the Middle Group Lavas of Piper (1977a) are now believed to record the segment of a transitional field path (D.N. Thomas, pers. commun.). Asymmetry of reversal is at least an incidental feature of the Upper Proterozoic geomagnetic field. It is recorded in the ~1100 Ma Mamainse Point lavas (Robertson, 1973) and in dyke swarms of the Thunder Bay district (Pesonen, 1979) of the Keweenawan igneous Province of North America. There is strong evidence from dyke contact zones in the latter example that the dykes record a primary TRM. It is unclear (cf. Palmer, 1970; Robertson, 1973 and Halls and Pesonen, 1983) whether more than one reversal is present in the Mamainse succession. The consistency of repeated asymmetry through eight field reversals in the Gardar succession and the component structure of the remanence (first subsection of third section, Pesonen, 1979 and Thomas and Piper, 1991) demonstrate that a secondary component compounded with a primary antiparallel axis is not an adequate explanation of asymmetry. Geomagnetic implications of this observation are discussed by Nevanelinna and Pesonen (1983). Because the younger units in the Gardar Province are magnetised predominantly with one polarity equivalent to the westerly declinations (it is not at present possible to identify unambiguously the normal or reversed sense of this magnetisation with respect to the present field) asymmetry cannot be recognised at later times in this area.

Shallow palaeolatitudes implied by the palaeomagnetic data from the Eriksfjord Formation decrease to equatorial values at the time recorded by the ~1290 Ma plutons (Fig. 10). When corrected for Mesozoic opening of the Labrador Sea between Greenland and North America (Table 6, Bullard et al., 1965) the poles are marginally removed from pole positions from the shield-wide dyke intrusion represented by the Mackenzie dyke swarm in central and northern parts of the Laurentian Shield and the Sudbury dykes at the southern margin

TABLE 6

Summary of palaeomagnetic results from the Gardar Igneous Province, South Greenland

Rock unit	Radiometric age (Ma)	Geologic age	Palaeomagnetic pole			Sites/ Samples	Palaeolatitude	Pole relative to Laurentian Shield		Reference
			E	N	dp/dm			E	N	
1. Ilímaussaq Complex										
(i) Fractionated rocks	1125±28	<(ii)	239	73	37/41	6/32	+60	241	76	Piper (1977b)
(ii) Syenites	1143±21	<unit 2	268	38	16/19	5/31	+54	256	38	" "
2. NE Dyke swarm	>1143, <1154	<unit 3	224	36	4/6	47/264	+32 (Tugtutôq)	210	40	Piper and Stearn (1977)
			231	33	5/7	18/102	+32 (Narssassuaqa)	217	37	This paper
3. Narssaq Gabbro and Ultramafics	>1143, <1154	<unit 5	225	32	8/12	4/22	+28	211	36	Piper (1977b)
4. Younger (gabbro) giant dykes, Tugtutôq	>1143, <1154	<unit 5	226	42	8/11	13/73	+37	213	46	" "
5. Older (Hviddal) giant dyke, Tugtutôq	1154± 9	<unit 10	215	33	8/12	7/38	+25	200	38	" "
6. South Qoroq Complex	1160± 8	>unit 2	214	42	5/7	9/67	+32	200	47	This paper
7. Gabbro Ring Dyke, Kungnat	1219±17	<unit 8	199	3	2/4	4/18	−7	182	9	Piper and Stearn (1977b)
8. WNW Dolerite Dykes		<unit 9, >unit 7	202	9	5/9	24/138	−1	185	15	" " "
9. NW Lamprophyre Dykes			206	3	5/10	9/51	−5	189	9	" " "
10. BDO Dykes		>units 1-5	227	11	8/14	2/14	+11	211	15	" " "
11. Narssassuaq Stock			207	9	4/7	3/23	0	191	15	This paper
12. North Qoroq Complex	1291±61	>unit 6	201	14	3/6	12/65	+1	185	20	" "
13. Motzfeldt Complex	1291±31	>unit 6	212	10	6/12	2/25	+2	196	15	" "
14. Eriksfjord Formation, Upper Lavas		>unit 13	222	16	5/9	24/113	(12)	206	21	Piper (1977a)
15. Eriksfjord Formation, Lower Lavas		>unit 14 (i)	239	7	7/14	7/34	(13)	223	10	This paper, basal lavas and carbonatite
		(ii)	207	10	7/14	5/36	(1)	190	16	
16. Eriksfjord Formation, Lower Redbeds		>unit 15	242	41	13/16	7/46	+43	230	44	Parsons (1990), late Gardar overprint

Units are listed in order of known or inferred increasing geological age. Palaeolatitude values in brackets are approximate due to inferred complexity of the geomagnetic field source during these times.

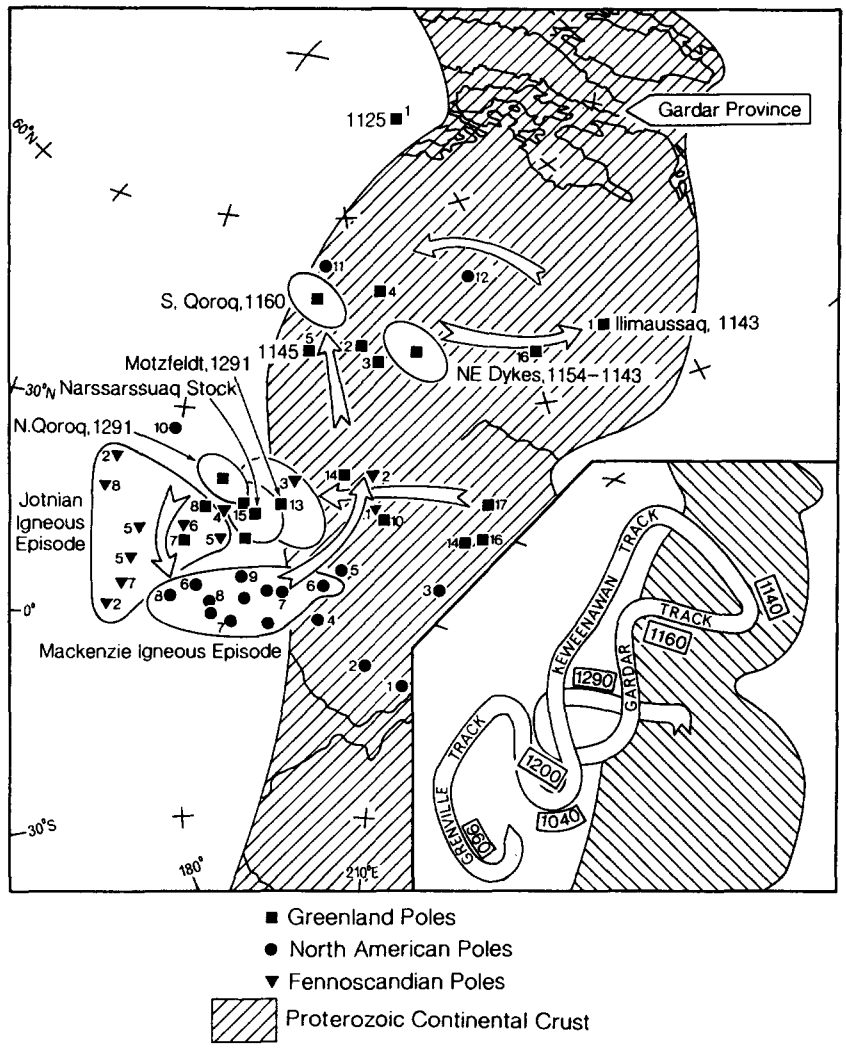


Fig. 11. Palaeomagnetic poles from the Gardar Province of South Greenland and selected contemporaneous poles (see: Piper, 1987, 1988) from the Laurentian Shield in North America and the Fennoscandian Shield. The Greenlandic poles (squares) are corrected with respect to Laurentia for the effects of Phanerozoic drift and are listed in Table 6. The poles from this study are shown with 95 percent confidence ovals. Two additional poles from elsewhere in Greenland are numbered with preferred ages in brackets: *16*, Zig-zag Dal basalts (< 1380 Ma) and *17*, Midsommersö dolerites (1250 Ma, probably same igneous event as 16). Dated North American poles (circles and see Piper 1988 for a summary) are numbered: *1*, Shabogamo gabbro north of Grenville Front (1375 Ma); *2*, Sherman granite and andesite dyke (1377 Ma); *3*, Mistastin pluton (1310 Ma); *4*, Big Springs granite borecore (1339 Ma); *5*, Seal Lake Group redbeds; *6*, Mackenzie dyke swarm (5 poles, 1251, 1232 Ma); *7*, Sudbury dykes and contacts (3 poles, 1205–1189 Ma); *8*, Coppermine Group lavas and sediments (2 poles, 1240 Ma); *9*, Muskox intrusion (1214 Ma); *10*, Gila County sills (1150 Ma); *11*, Lackner Lake Complex (1095 Ma); *12*, Logan Sills (1109 Ma). Fennoscandian poles (triangles) are numbered: *1*, St Tuna dolerites (1371 Ma); *2*, Sundsjö-giman dolerite dykes (3 poles, 1229–1156 Ma); *3*, Alvhö dolerite sills (1215 Ma); *4*, Vaasa dolerite dykes (1270–11225 Ma); *5*, Nordingrå dolerites and overprint (3 poles, 1245 Ma); *6*, Market dolerite dykes (1270 Ma); *7*, Gnarp dolerites (1245 Ma); *8*, Alvdalsan dolerite sills (1231 Ma). The Fennoscandian poles are rotated into the Proterozoic quasi-rigid configuration with North America retained in present day coordinates; the Lambert equal area projection is sectored in 30 divisions. The integrity of this larger area of crust [in palaeomagnetic terms, see Piper (1987) for rotation poles] is required by the single peak distribution property of Precambrian poles from all shields which is satisfied by no other tectonic premise (Piper, 1991). The inset figure summarises the APW track defined from the Gardar Province; the succeeding track (ca. 1100–1050 Ma) is defined from the Keweenawan Province of North America. See Piper (1987) for further details.

(Fig. 11 and Piper, 1987). In the Fennoscandian Shield the post-Jotnian igneous episode in Sweden and Finland defines a group of poles bordering pole positions from the Laurentian Shield on the pre-1100 Ma reconstruction of the two shields (Patchett et al., 1978; Stearn and Piper, 1984). The Mackenzie, Sudbury and related events have age assignments predominantly in the range 1250–1200 Ma and the bulk of the ages from the post-Jotnian episode fall in the age range 1270–1215 Ma (Patchett, 1978). The Mackenzie and post-Jotnian rocks all have the same polarity equivalent to the predominant polarity in the slightly-older Gardar rocks. Hence, following the interval of high reversal frequency recorded in the Eriksfjord Formation, the geomagnetic field appears to have been characterised by a long period ($\sim$1290–1200 Ma) of constant polarity. During these times rates of APW were also low. They are embraced by a loop no more than 20–30° in amplitude (Fig. 11).

A polar shift, but the same field polarity, is recorded in later (1160–1145 Ma) members of the Gardar Province (Table 6) including the South Qoroq complex and the NE–SW trending dyke swarm. A continuation of this motion appears to be present in the youngest members of the Gardar Province (Fig. 11 and Table 6) but is at present poorly-defined. Red sandstones interleaved with the lavas of the Eriksfjord Formation are variably bleached along later NE–SW trending Gardar fractures and in dyke margins of this generation, and it is likely that their diagenesis was not completed until late in the Gardar period. These rocks have a characteristic magnetisation (Parsons, 1990 and result 16 in Table 6) similar to the direction of remanence in the South Qoroq complex and the majority of the NE–SW trending dykes. Hence this remanence is regarded as a late Gardar ($\sim$1140 Ma) acquisition.

The earliest phase of magmatism in the Gardar Province is approximately contemporaneous with basaltic volcanism in North Greenland. Palaeomagnetic studies of these rocks are reported by Marcussen and Abrahamsen (1983) and Abrahamsen and Van der Voo (1987). The pole positions (Midsommersö dolerites 242°E, 6.9°S; Zig Zag Dal basalts 242.8°E, 12.2°S; Victoria Fjord nunatak 231.7°E, 10.3°N) lie to the east of the $\sim$1290 Ma poles from the Gardar Province but correspond to the earliest poles defined by the Eriksfjord Formation and possibly to the $\sim$1250 Ma BDO dykes (Table 6) although the latter are currently represented by only two sites. The results from Victoria Fjord nunatak is of opposite polarity to the other two results from north Greenland. Unfortunately the studied units in this area are not directly linked to age determinations; they postdate $\sim$1380 Ma sediments and analogous basalts in the Independence Fjord yield Rb–Sr isochrons of $\sim$1250 Ma (Marcussen and Abrahamsen, 1983). Hence they may correlate either with the initial part of the Gardar Track or with the cross-over point of the tracks in Fig. 11. If the latter is the case, it will show the interval 1290–1210 Ma was not characterised exclusively by one polarity.

Thus the collective data from the Gardar Province define a small APW loop followed by a long track. The latter is contemporaneous with the outward path of the "Great Logan Loop" recognised initially by DuBois (1962) and subsequently refined by Robertson and Fahrig (1973). The status of this loop has however, undergone major revision since the early 1970's. The results from the Keweenawan Province are now considered to define only the return limb of the loop (Halls and Pesonen, 1983; Palmer and Halls, 1986). This latter path may also have been rapidly executed: it traverses about 60° of arc and embraces two segments of opposite polarity. Palmer and Davis (1987) consider that the bulk of the Keweenawan igneous activity occurred over an interval of $\sim$22 Ma at $\sim$1090 Ma. Definition of the amplitude of the APW loop is dependent on the geometry of the geomagnetic field source during these times (Neva-

nelinna and Pesonen, 1983). However, the earliest segment of this track is defined by poles of both polarity from alkaline bodies in the Laurentian Shield. The key dated poles at the beginning of this path come from the Logan Sills dated by high precision U–Pb methods at 1109 Ma (12 in Fig. 11) and the Lackner Lake Complex dated by Rb–Sr at 1095 Ma (pole 11 in Fig. 11, Palmer and Davis, 1987 and Symons, 1989a,b). These poles lie at the apex of the loop defined in the inset of Fig. 11 and close to the youngest (1145–1125 Ma) segment of the APW path defined by the Gardar results. There is as yet no positive evidence that asymmetry characterised the time represented by this apex although it is a feature of earlier (early Gardar) and later (Keweenawan) times. Since the ~1100 Ma poles from the Laurentian Shield plot in the vicinity of youngest poles from the Gardar Province. The "Keweenawan Track" is identified as the return path of an APW loop. The outward path is defined by the sequence of magmatic events in the Gardar Province (Table 6 and Fig. 11) and is designated the "Gardar Track".

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